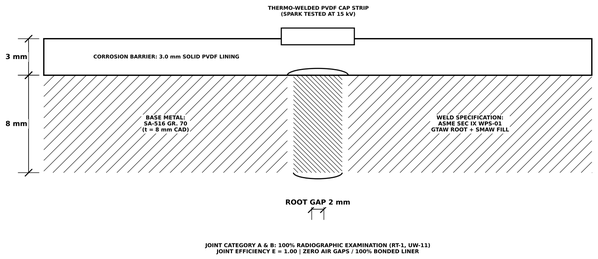
DETAIL 1 — ASME SEC VIII DIV 1 SHELL BUTT WELD & LINER

TYPE 1 FULL-PENETRATION DOUBLE-WELDED JOINT & PVDF BARRIER (SCALE 1:2 METRIC)



JOINT CATEGORY A & B: 100% RADIOGRAPHIC EXAMINATION (RT-1, UW-11)
JOINT EFFICIENCY E = 1.00 | ZERO AIR GAPS / 100% BONDED LINER

DETAIL 2 — NOZZLE N1 (DN150) PENETRATION & REINFORCING PAD

ASME B16.5 WELD NECK FLANGE & ASME UG-37 REPAD (SCALE 1:5 METRIC)

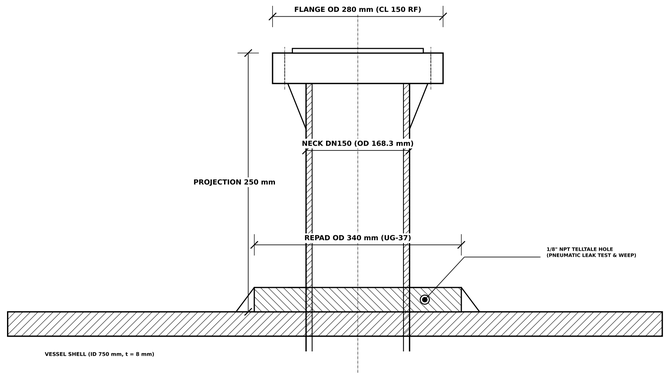
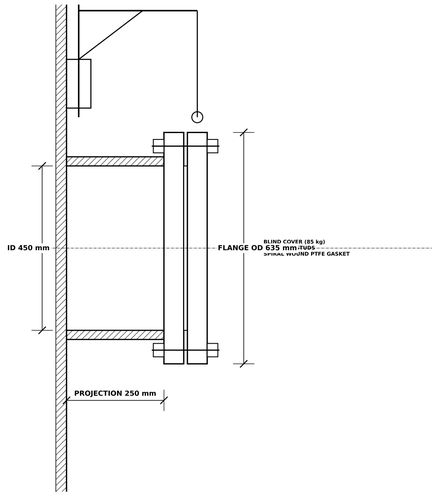


TABLE 1 — MASTER ASME B16.5 NOZZLE & REINFORCEMENT SCHEDULE

TAG	SERVICE / DESCRIPTION	SIZE	RATING	FLG OD	BOLTS	REPAD
N1	Feed Gas Inlet Mixture	DN 150 (6")	CL 150	280 mm	8x M20x110	Ø340 mm
N2	Gas Sampling Connection	DN 40 (1.5")	CL 150	127 mm	4x M14x75	None
N3	Scrubbing Water Feed Inlet	DN 80 (3")	CL 150	190 mm	4x M16x90	Ø190 mm
N4	Rich Liquor Sump Outlet	DN 80 (3")	CL 150	190 mm	4x M16x90	Integral
N5	Column Shell Bottom Drain	DN 40 (1.5")	CL 150	127 mm	4x M14x75	None
N6	Treated Overhead Gas Vent	DN 50 (2")	CL 150	152 mm	4x M16x85	None
N7	Bed Differential Press (Lo)	DN 50 (2")	CL 150	152 mm	4x M16x85	None
N8	Bed Differential Press (Hi)	DN 50 (2")	CL 150	152 mm	4x M16x85	None
N9	Gas Inlet Thermowell (TI)	DN 40 (1.5")	CL 150	127 mm	4x M14x75	None
N10	Liquid Level Instrument Taps	DN 50 (2")	CL 150	152 mm	4x M16x85	None
N11	Demister Flush Spray Line	DN 40 (1.5")	CL 150	127 mm	4x M14x75	None
M1	Lower Packing Access Manway	DN 450 (18")	CL 150	635 mm	16x 1-1/8"	Ø750 mm
M2	Upper Distributor Manway	DN 450 (18")	CL 150	635 mm	16x 1-1/8"	Ø750 mm

DETAIL 3 — VESSEL ACCESS MANWAY (DN450) & DAVIT ARM

ASME B16.5 CLASS 150 BLIND COVER, SPIRAL GASKET & HINGED DAVIT (SCALE 1:6)



DETAIL 4 — TOP LIFTING LUG & EARTHING BOSS DETAILS

RIGGING PADEYE (5.0 t SWL) & GROUNDING TERMINATION (SCALE 1:5 METRIC)

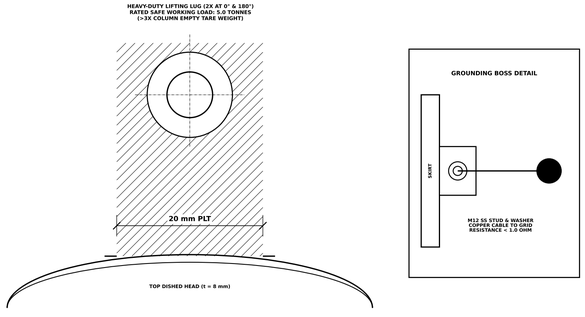


TABLE 2 — WELDING PROCEDURE (ASME SEC IX) & NDE REQUIREMENTS

JOINT CAT.	SERVICE / LOCATION	WELD TYPE	PROCESS	NDE / INSPECTION
Cat. A	Longitudinal Shell Seams	Double-V Butt	GTAW + SMAW	100% Radiography (RT-1)
Cat. B	Circumferential Girth Seams	Double-V Butt	GTAW + SMAW	100% Radiography (RT-1)
Cat. C	Flange-to-Neck Welds	Full Pen Hub	GTAW + SMAW	100% Dye Penetrant (PT)
Cat. D	Nozzle-to-Shell Welds	Full Penetration	GTAW + SMAW	100% PT + Ultrasonic (UT)
Repad	Reinforcing Pad Fillet	Continuous Fillet	SMAW E7018	Air Leak Test (1.5 bar)
Liner	PVDF Corrosion Barrier	Hot Gas Thermo	Thermoplastic	Spark Test 15 kV (NACE)

PRESSURE VESSEL FABRICATION & TESTING SPECIFICATIONS

1. ASME CODE JURISDICTION & DESIGN MARGINS:

- FABRICATED IN STRICT ACCORDANCE WITH ASME SECTION VIII DIVISION 1 (2023 EDITION).
- MAXIMUM ALLOWABLE WORKING PRESSURE (MAWP): 150 kPa (g) + FULL VACUUM CONTINGENCY.
- MINIMUM DESIGN METAL TEMPERATURE (MDMT): -10 °C AT DESIGN PRESSURE.

2. CORROSION LINING CONTINUITY & SPARK TESTING:

- **ALL PROCESS WETTED INTERIOR SURFACES SHALL BE LINED WITH 3.0 mm BONDED PVDF / PTFE.**
- LINER CONTINUITY TESTED OVER 100% OF SURFACE USING HIGH-VOLTAGE SPARK TESTER AT 15 kV.
- ZERO POROSITY, BLISTERS OR AIR GAPS ALLOWED BETWEEN STEEL SHELL AND FLUOROPOLYMER LINER.

3. PRESSURE TESTING & POST-WELD HEAT TREATMENT (PWHT):

- **HYDROSTATIC TEST CONDUCTED AT 1.5x MAWP PRIOR TO FLUOROPOLYMER LINING INSTALLATION.**
- TEST WATER: CHLORIDE-FREE DEMINERALIZED WATER (< 25 ppm Cl-), TEMPERATURE >= 20 °C.

THIRD ANGLE PROJECTION

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE **PACKED-BED ABSORBER (C-101) — MECHANICAL FABRICATION, NOZZLE & MANWAY DETAILS CAD**	LAB / SUBJECT **PROCESS EQUIPMENT DESIGN & DRAWING**	COURSE / BRANCH **B.E. CHEMICAL ENGINEERING**	SCALE & REVISION **SCALE: DETAILED VIEWS 1:2, 1:5, 1:6 METRIC | REV: 00**
STUDENT NAME **SOURISH SENAPATI**	EXAM ROLL NUMBER **002310301035**	DRAWING / DOCUMENT NO. **PEDD-PBA-04 (SHEET 04 OF 05)**	DATE **29 / 09 / 2026**
EQUIPMENT IDENTIFICATION & SERVICE **PRESSURE VESSEL FABRICATION DETAILS — WELDS, REPAD, MANWAY & RIGGING**	DESIGN BASIS & METALLURGY / LINING **ASME SEC VIII DIV 1 / SA-516 GR 70 + PVDF LINER / SS316L / CERAMIC**	PROCESS PERFORMANCE & RECOVERY **ASME SEC VIII DIV 1 | SA-516 GR 70 + PVDF LINER | ASME B16.5 CL 150 RF**	DESIGN STATUS CLASSIFICATION **CALCULATED: ID 750, Z 4500 | PRELIMINARY CAD: SHELL & NOZZLES**

DIMENSIONS: METRIC (MM / M) | FLANGES: ASME B16.5 CLASS 150 RF | WELDING: ASME SEC IX | THIRD ANGLE PROJECTION